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Student Number: 2986732C	Student Name: Zhuofan Cui
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Name of First Supervisor: Olaoluwa Popoola	Name of Second Supervisor: Heba Dawoud
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Flexible Physical Layer Based Resource Allocation for Mission-Critical Communications in 6G

Zhuofan Cui

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Abstract

As 6G smart factories accommodate a massive number of diverse devices, co-scheduling URLLC and mMTC services on shared physical resources emerges as a critical challenge. This thesis proposes a resource allocation mechanism for this convergence scenario, combining flexible numerology configuration and priority control at the physical layer. By formulating a multi-objective optimization model that integrates delay constraints, channel conditions, and spectral efficiency, we design a low-complexity heuristic scheduling algorithm. Furthermore, an SNR enhancement strategy is incorporated to boost the schedulability of URLLC devices in complex environments. Simulation results indicate that our proposed method significantly enhances resource utilization while ensuring URLLC reliability, improving system schedulability by approximately 20.6%. This project delivers effective technical support for establishing industrial wireless communication systems capable of coexisting multi-QoS services.

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Abbreviations

IoT	Internet of Things
AGVs	Automated guided vehicles
mMTC	Massive Machine-Type Communications
QoS	Quality of Service
URLLC	Ultra-Reliable Low-Latency Communications
IMT	International Mobile Telecommunications
ITU	International Telecommunication Union
BWP	Bandwidth Parts
MILP	Mixed-Integer Linear Programming
eMBB	Enhanced Mobile Broadband
NLoS	Non-Line-of-Sight
RB	Resource Block
SNR	Signal-to-Noise Ratio
IIoT	Industrial Internet of Things
BS	Base station
gNB	gNodeB
InF-SL	Indoor Factory Sparse Clutter Low Base Station Height
LoS	Line-of-sight
3GPP	3rd Generation Partnership Project
PL	Path Loss
MINLP	Mixed-Integer Nonlinear Programming
AWGN	Additive White Gaussian Noise

Table of Contents

Abstract	i
Acknowledgments	ii
Abbreviations	iii
Table of Contents	iv
1. Introduction	1
1.1 Background	1
1.2 Aim and Objectives	3
1.3 Contributions	3
1.4 Outline	4
2. Literature Review	5
2.1 Overview of 6G architecture relevant to URLLC/mMTC	5
2.2 Related Work	5
2.3 Research Gaps	6
3. Methodology	8
3.1 System Model	8
3.1.1 System Architecture	8
3.1.2 Channel Model	9
3.1.3 Flexible Time—frequency resource structure in 5G NR	11
3.2 Problem formulation	13
3.3 Solution to the Multi-Objective Optimization Problem	14
3.3.1 Initialization Phase	15
3.3.2 SNR Estimation and Boosting Phase	16
3.3.3 Scheduling phase	17
4. Results and discussion	19
4.1 Simulation Setup	19
4.2 Simulation Results	20
4.2.1 Device Deployment and LoS/NLoS Classification	20
4.2.2 SNR Distribution before/after Boosting	21
4.2.3 URLLC Device Latency Distribution	22
4.2.4 URLLC Latency CDF Comparison	24
4.2.5 Resource Distribution Results	25
4.2.6 Resource Utilization of URLLC and mMTC Device	27
4.2.7 SNR Boosting Improvement per MC Run	29
4.2.8 URLLC Latency Evolution Across 40 Monte Carlo Runs	31
4.2.9 Service Type Resource Utilization Comparison with Trends	32
5. Conclusion	34
6. Reference	35

1. Introduction

1.1 Background

The fourth industrial revolution (Industry 4.0) is characterized by the integration of cutting-edge digital technologies into manufacturing to increase productivity and flexibility. A key outcome is the emergence of smart factories, which leverage ubiquitous connectivity and automation to optimize production processes in real time. With the accelerated advancement of Industry 4.0 and smart manufacturing, smart factories are gradually transitioning from traditional wired control architectures to network-based control systems characterized by ubiquitous wireless connectivity [1]. By deploying networks of Internet of Things (IoT) sensors and devices for real-time monitoring and control, manufacturers can achieve predictive maintenance, energy optimization, and enhanced safety in operations. This shift from fixed wired infrastructure to flexible wireless networking forms the digital nervous system of the future factory [2].

However, this transition to wireless-based smart factories introduces significant communication challenges. As indicated by existing research, future factories will deploy thousands of heterogeneous end devices, including sensors, collaborative robots, and automated guided vehicles (AGVs) [2] [3]. These devices exhibit fundamentally different communication requirements: certain non-critical devices (such as environmental sensors and electricity meters) tolerate higher latency and are better suited for Massive Machine-Type Communications (mMTC) connectivity. In contrast, devices involved in real-time closed-loop control, such as robotic arm drives, have stringent end-to-end latency and reliability requirements and necessitate Quality of Service (QoS) guarantees at the Ultra-Reliable Low-Latency Communications (URLLC) level [4]. Consequently, accommodating both categories in the same network means that the infrastructure must handle simultaneously, extremely time-sensitive traffic alongside massive-volume device connections. This highly hybrid service scenario poses a significant challenge to traditional cellular communication architectures, which were not originally designed to meet such concurrent extremes in reliability and low latency. Conventional approaches, such as simple network slicing that isolates mMTC from URLLC, are insufficient because they cannot simultaneously guarantee all required performance metrics within a unified system. The resulting challenge is how to maintain ultra-low latency for mission-critical flows without sacrificing the capacity to support massive device density.

While the current 5G infrastructure is insufficient to fully support these stringent

industrial communication needs, emerging 6G technologies offer promising solutions to bridge this gap. In the envisioned 6G era, the wireless factory network evolved from an auxiliary support system to become the "neural hub" carrying mission-critical control and information flows at the core of production. According to the International Mobile Telecommunications (IMT)-2030 vision proposed by the International Telecommunication Union (ITU) [5], 6G is expected to support sub-millisecond end-to-end latencies (on the order of 0.1 ms), reliability levels up to 10^{-7} for critical links, and a connectivity density of millions of devices per square kilometer. These ambitious requirements impose extremely high demands on network resource allocation efficiency, flexibility, and service differentiation capabilities. In addition, many factory devices also have different needs in terms of jitter, timing constraints, response speed, and other QoS requirements [6]-[10]. As 6G research advances, a key scientific problem has emerged: how to simultaneously satisfy the constraints of URLLC and mMTC devices under limited physical resources. In other words, how to deliver ultra-high reliability and sub-1ms responsiveness for some devices while hundreds of other devices share the same resource. Solving this problem is critical for developing next-generation smart factory communication systems [11].

To address the coexistence challenge, recent standardization efforts have introduced several promising technical approaches. Mixed-numerology and bandwidth parts (BWP) have been proposed to accommodate heterogeneous industrial traffic on a shared carrier, allowing subcarrier spacing, symbol duration, and effective TTI to be tailored per service; latency-critical control loops can operate with short symbols/mini-slots while delay-tolerant devices use more spectrally efficient settings on the same spectrum. However, this flexibility introduces coordination challenges such as guard overhead and inter-numerology interference. Without proper management, these issues can significantly degrade the benefits of coexistence. In parallel, QoS-constrained scheduling has been identified as essential for protecting URLLC deadlines and reliability when resources are multiplexed with other traffic classes [12]. Addressing these technical challenges forms the foundation of current research efforts. Ultimately, the goal is to develop communication strategies that guarantee URLLC-level low latency and high reliability in a crowded industrial wireless environment, while still efficiently supporting massive-scale IoT connectivity. This objective directly serves the industry 4.0 vision by enabling wireless networks to meet or exceed the performance once only achievable with wired industrial systems, thus unlocking the full potential of future smart factories.

1.2 Aim and Objectives

i. General objectives:

Designing an Efficient Resource Allocation Algorithm to Solve the Bandwidth Resource Allocation Problem for URLLC and mMTC Devices in 6G Smart Factory

ii. Specific Objectives

- A. Develop a Monte Carlo-based simulation model for device deployment, channel classification (LOS/NLOS), and system parameter modeling in a factory scenario.
- B. Apply distance-based LOS probability and Path Loss (PL) models to compute realistic SNR values for URLLC and mMTC devices under different propagation conditions.
- C. Evaluate the effectiveness of the proposed algorithm in terms of resource allocation efficiency, bandwidth utilization, and latency performance across multiple Monte Carlo runs.

1.3 Contributions

This thesis proposes a QoS-focused resource allocation mechanism for 6G smart factories, optimizing collaboration between URLLC and mMTC devices with shared resources. The main contributions of this work can be summarized as follows:

- A. We propose a novel Signal-to-Noise Ratio (SNR) boosting strategy that selectively enhances the transmit power for critical URLLC devices under Non-Line-of-Sight (NLoS) conditions. This mechanism enhances the capacity to schedule more high-priority devices and ensures reliability under challenging channel environments.
- B. We formulate a multi-objective resource allocation problem for 6G industrial networks that support both URLLC and mMTC devices. The goal is to minimize the overall system bandwidth consumption and the average communication latency, while ensuring that each device is assigned exactly one Resource Block (RB) and meets its QoS delay constraints.

- C. We design a context-aware scheduling framework that integrates SNR-aware filtering, numerology mapping, and priority-based allocation. We introduce a weighted optimization model with adjustable parameters that balance delay performance and spectrum efficiency under heterogeneous QoS constraints. The proposed adaptive algorithm accounts for service-criticality, network density, and dynamic channel conditions, enabling scalable and reliable resource allocation.
- D. We evaluate the performance of our framework through simulation, demonstrating that our SNR-enhanced scheduling model improves URLLC reliability, reduces average latency, and achieves higher bandwidth efficiency compared to existing baseline strategies.

1.4 Outline

Section 1 discusses the coexistence challenge between URLLC and mMTC devices in Industry 4.0 smart factories. The background covers the transition from wired to wireless control systems and the stringent performance requirements for mission-critical applications. The core problem of simultaneously achieving sub-millisecond latency for control loops and massive device connectivity is formulated, establishing objectives for developing resource allocation and scheduling strategies in mixed-numerology systems that guarantee both URLLC reliability and mMTC efficiency. Section 2 reviews recent advances in resource allocation for URLLC in smart factory scenarios, highlighting emerging technologies and novel algorithms proposed in the literature. Based on this survey, several research gaps are identified, which motivate the present work and lead to the formulation of our research contributions. The scenario and system model are established in Section 3, where channel modeling and the flexible time–frequency structure are presented. The problem formulation and the proposed scheduling algorithm are also presented in Section 3. Section 4 presents the performance set-up and discusses the results. It first introduces the simulation environment, including system and channel parameters. Then, it analyzes key performance metrics to evaluate the proposed algorithm's effectiveness under diverse QoS and channel conditions. Finally, the conclusion is given in Section 5.

2. Literature Review

In this section, we provide an overview of emerging 6G network architectures with a focus on enabling ultra-reliable low-latency communication (URLLC) and massive machine-type communication (mMTC) service classes. Then we summarize the state-of-the-art methods of existing resource allocation and their limitations.

2.1 Overview of 6G architecture relevant to URLLC/mMTC.

The scope of our review spans key technologies and design principles proposed to meet the stringent requirements of URLLC alongside the massive connectivity of mMTC. 5G networks introduced three broad service categories eMBB, URLLC and mMTC – each with distinct performance targets. Traditionally, these services have been provisioned in isolation (e.g., via dedicated slices). However, the evolution toward 6G is driven by the need to support heterogeneous service classes concurrently on a unified platform. Emerging industrial and IoT applications often demand a combination of previously separate 5G capabilities. For instance, smart factory deployment requires both ultra-low latency and high reliability for real-time control (URLLC) and massive connectivity for thousands of wireless sensors (mMTC). Recent studies have even defined integrated service notions like critical mMTC, which blend URLLC-level reliability/latency with mMTC-scale connectivity in one category [13]-[15]. Such scenarios reveal the limitations of simple 5G-era solutions – a conventional mMTC–URLLC mixture via network slicing cannot simultaneously guarantee sub-5 ms end-to-end latency and million-device scalability under the same network conditions. This recognition has underscored the importance of new architectural frameworks beyond 5G that natively accommodate multi-service ultra-critical requirements.

2.2 Related Work

In recent years, significant advancements have been made in resource allocation strategies for URLLC in smart factories. For instance, one study established a detailed downlink URLLC system model for a smart factory under the 5G NR standard, formulating a joint time–frequency resource and power allocation problem under strict delay and reliability constraints. By decomposing this challenging mixed-integer optimization, the authors applied a matching game (solved via a deferred acceptance algorithm) for scheduling and a water-filling power allocation strategy, effectively maximizing network throughput while meeting URLLC's QoS targets. As a result, the proposed scheme achieved a notable increase in system throughput compared to baseline allocations [16]. Building on this foundation, other research explores the Open-

RAN architecture and develops a dynamic resource allocation framework. This framework integrates a Transformer network with a Mixed-Integer Linear Programming (MILP) model, enabling highly timely scheduling decisions that accommodate diverse service QoS constraints. Furthermore, it demonstrates the performance advantages of the Flexible Numerology mechanism in low-latency transmission scenarios [17]. Meanwhile, for mMTC devices, a semi-persistent resource allocation algorithm based on a multi-numerology structure has been proposed to improve scalability. This scheme clusters MTC devices with similar QoS requirements and assigns them appropriate numerologies such that multiple device groups can be served concurrently with minimal scheduling overhead. By employing spectrum division and a low-complexity clustering optimization mechanism, the algorithm significantly enhances resource reuse efficiency and connection density without increasing signaling overhead, demonstrating strong potential for 6G networks to support massive IoT connectivity [18]. Additionally, as a significant evolution of the 5G NR physical layer, the introduction of flexible multi-numerology systems (with configurable subcarrier spacing, symbol duration, and subframe structure) provides critical support for diversified industrial scenarios. This design allows the network to tailor physical-layer parameters for different applications. For example, using a larger subcarrier spacing (and thus shorter transmission time interval) to achieve lower latency for URLLC, or a narrower spacing to improve coverage and efficiency for mMTC. Such flexibility establishes a foundation for future resource allocation mechanisms in 6G, enabling schedulers to dynamically adjust the frame configuration to meet each service's unique QoS demands [19] [20]. Moreover, researchers have introduced multi-connectivity as another key technique to further bolster URLLC reliability. In practice, this approach often involves duplicating URLLC packets across multiple base stations (e.g. via dual connectivity) and transmitting them simultaneously over independent links. Coordinating these parallel transmissions – even if it means preempting some Enhanced Mobile Broadband (eMBB) resources – substantially increases the probability of meeting URLLC's stringent block error rate targets, since at least one of the duplicate packets can reach the receiver on time. At the same time, intelligent scheduling is used to limit such duplication to minimize the impact on coexisting eMBB throughput, thereby preserving overall network efficiency [21].

2.3 Research Gaps

However, existing research still exhibits several key limitations. Firstly, regarding flexible numerology resource allocation, most approaches adopt simplified or idealized channel models that fail to capture the complex and harsh characteristics of industrial wireless environments. These models often ignore factors like frequent obstructions by moving machinery, rich multipath

reflections from metal surfaces, and high-density device deployments. Such oversimplifications lead to results that may not hold up under real factory conditions, weakening the practical applicability of the proposed schemes [17] [18]. Secondly, most resource allocation optimizations for URLLC traffic still rely on fixed-numerology designs, without integrating the flexible multi-numerology mechanisms enabled in 5G and beyond. In other words, they use fixed subcarrier spacing and slot duration in advance, lacking any mechanism to dynamically adjust numerology to varying QoS demands. This gap means current scheduling frameworks cannot exploit numerology flexibility to better satisfy diverse and dynamic latency or reliability constraints, making them less adaptable to changing service requirements [16]. Thirdly, current research predominantly focuses on single-service scenarios (either URLLC or mMTC alone), with very limited exploration of coexistence mechanisms or scheduling conflict resolution strategies when URLLC and mMTC traffic share the same physical resources. In practice, treating these services in isolation has left challenges like cross-service interference, resource contention, and priority inversions largely unaddressed. This makes it difficult to guarantee stringent URLLC QoS while simultaneously supporting massive IoT connectivity, thereby hindering the realization of multi-service heterogeneous concurrency demanded in future 6G smart factories [16] [18] [20].

Paper	URLLC Support	mMTC Support	SNR Boosting	QoS Class Mapping	Channel Dynamics Considered	Multi-Numerology
[16]	√				√	
[17]	√					√
[18]		√				√
[19]	√	√				√
[20]	√				√	√
This thesis	√	√	√	√	√	√

Table 1: Comparative Summary of Related Work

According to the Tabel 1, this project aims to design a physical layer resource allocation mechanism oriented toward QoS for typical service convergence scenarios in 6G smart factories. The goal is to achieve efficient collaboration between URLLC and mMTC devices under shared resources.

3. Methodology

3.1 System Model

To rigorously evaluate how URLLC and mMTC services can coexist over shared spectrum in future smart factories, a well-defined system model is essential. This includes realistic characterization of the wireless environment, as industrial deployments often feature dense device distributions, frequent obstructions, and NLOS conditions that significantly affect signal quality and resource schedulability. Accurate channel modeling enables reliable assessment of SNR-driven QoS satisfaction, which directly impacts resource allocation decisions. Moreover, to support service heterogeneity under strict latency and reliability constraints, the system must incorporate the flexible time–frequency structure introduced in 5G NR, such as multiple numerologies, mini-slots, and bandwidth parts, which allows for differentiated transmission durations and spectrum granularity. These structural elements form the foundation upon which adaptive, QoS-aware scheduling mechanisms can be built.

3.1.1 System Architecture

In this project, we consider a typical Industrial Internet of Things (IIoT) scenario where a single 5G base station is deployed to manage efficient resource allocation for multiple heterogeneous devices within a 50×50 m² factory. These devices exhibit diverse operational requirements, posing significant resource scheduling challenges.

As shown in Figure 1, a Medium Range Base station (BS), represented as gNodeB (gNB), is placed approximately 10 meters outside the factory to provide comprehensive coverage [22]. We assume there are $I=100$ devices, denoted by the set $\{i_1, i_2, i_3, \dots, i_N\}$, randomly distributed throughout the factory. Each device $i_n \in I$ is equipped with a single antenna positioned at approximately height $h_d = 1.5$ m and connects to the base station BS for downlink control and data transmission. This project focuses on downlink communications within this single-cell network, while uplink communication and external cell interference are not considered due to the isolated nature of the factory scenario.

The devices in this scenario fall into two primary QoS service classes and differ in terms of reliability and delay:

- i. URLLC devices: A subset $I_{URLLC} \subset I$, including devices with stringent latency and high reliability requirements, such as robotic arms and emergency stop buttons. These devices are given high scheduling priority.
- ii. mMTC devices: A subset $I_{mMTC} = I - I_{URLLC}$, consisting of low

power meters and humidity sensors [4]. Although these devices have less stringent latency requirements ($T_{mMTC} > T_{URLLC}$), their massive numbers result in significant demands for spectrum resources.

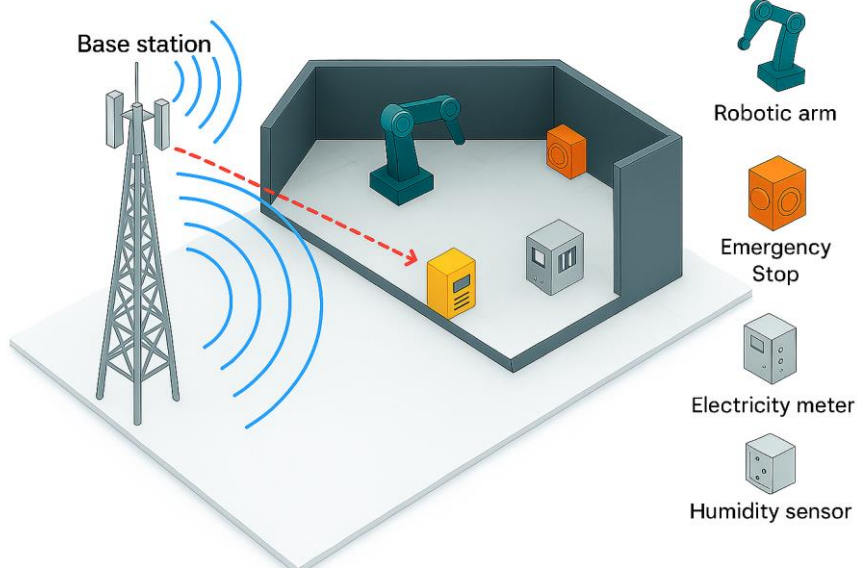


Figure 1: System Architecture

3.1.2 Channel Model

In this project, we conduct channel modeling for static industrial equipment communication within suburban factory environments, based on the Indoor Factory Sparse Clutter Low Base Station Height (InF-SL) scenario model as specified in 3rd Generation Partnership Project (3GPP) TR 38.901 with a center frequency $f_c = 3.5\text{GHz}$ [23] [24]. This model is characterized by sparsely distributed obstacles within the factory, such as machinery, equipment, and production lines, with both the base station and terminal antenna heights being lower than the average clutter height [23]. Given the relatively stationary nature of industrial equipment and the base station, we adopt a quasi-static channel model [16]. This model ignores the effects of small-scale fast fading and considers only large-scale fading, which includes PL and lognormal shadow fading. Under this assumption, the channel remains constant during transmission, with variations in the received signal primarily attributed to slow changes from PL and shadowing.

A. LoS Probability Model:

For the described scenario, 3GPP specifies a distance-dependent LoS probability model. Let d_{2D} represents the distance between the base station gNB and the device d_i . The probability of the device being in the LOS state at this distance can be expressed in an exponentially decaying for [23]:

$$\Pr_{\text{LOS,subsc}}(d_{2D}) = \exp\left(-\frac{d_{2D}}{k_{\text{subsc}}}\right) \quad (1)$$

The parameter $k_{\text{subsc}} = -\frac{d_{\text{clutter}}}{\ln(1-r)}$ is the environmental attenuation distance, which is determined by the plant layout parameters, derived from the average obstacle spacing d_{clutter} and the empirical value r . The LoS probability model is based on the average distance between obstacles and illustrates the rapid decline in the likelihood of LoS occurrence as the distance increases.

B. Propagation Distance Modeling:

In accordance with the 3GPP model, we separate the total propagation distance into two segments in this thesis: the outdoor and indoor parts, given the existence of a propagation path from outdoor to indoor. The total 3D propagation distance $d_{3D,\text{total}}$ is calculated as [23]:

$$d_{3D,\text{total}} = \sqrt{(d_{2D,\text{out}} + d_{2D,\text{in}})^2 + (h_{BS} - h_{UT})^2} \quad (2)$$

where $d_{2D,\text{out}}$ and $d_{2D,\text{in}}$ are the horizontal distances from the base station to the exterior wall of the building and the horizontal distances of the signal propagation in the interior, respectively. h_{BS} and h_{UT} are the heights of the base station and antenna, respectively. This formula synthesises the outdoor and indoor transmission paths into an equivalent 3D propagation distance for the purpose of the PL calculation, as discussed in the next subsequent section.

C. PL Models:

In industrial scenarios such as InF-SL, different PL (PL) models are required for different propagation conditions. For the LoS condition, the PL is usually modelled as the logarithm of distance and center frequency. Therefore, the LoS PL model for InF-SL environment at center frequency f_c (GHz) can be expressed as [23]:

$$PL_{\text{LOS}} = 31.84 + 21.50 \log_{10}(d_{3D}) + 19.00 \log_{10}(f_c) \quad (3)$$

However, in the NLoS scenario, there is a high density of clutter inside the plant, and much of the equipment is obscured by objects such as shelves and walls. Therefore, the NLoS PL is modelled as [23]:

$$PL_{\text{NLOS}}^{\text{InF-SL}} = 33 + 25.5 \log_{10}(d_{3D}) + 20 \log_{10}(f_c) \quad (4)$$

$$PL_{\text{NLOS}} = \max(PL, PL_{\text{LOS}}) \quad (5)$$

It shows that the distance attenuation factor is greater in the NLoS case, reflecting the severe additional attenuation in dense clutter environments.

D. SNR Calculation:

Based on the above large-scale fading channel model, the SNR of each device is obtained by modelling using the discriminative mechanism $\frac{Pr}{LOS,subsc}$. The

SNR of a single link can be determined from the transceiver parameters and PL. The base station transmitter power is denoted as P_{tx} , the transmit and receive antenna gains are G_{tx} and G_{rx} , respectively, and the PL after inclusion of shadowing fading is PL . The receiver noise power is N_0 , which can be calculated from the Boltzmann constant [24]. The SNR is defined as the ratio of the signal power to the noise power. Therefore, the SNR is given by:

$$SNR_i = P_{tx} + G_{tx} + G_{rx} - PL - N_0 \quad (6)$$

The primary role of the SNR in this work is to quantify the wireless link quality of each device, serving as a key criterion for schedulability determination.

3.1.3 Flexible Time—frequency resource structure in 5G NR

To meet the diverse QoS requirements, particularly for URLLC applications in industrial scenarios, all devices and the base station operate on a total available bandwidth of $B_{tot} = 100$ MHz, assumed to lie in a licensed 6G industrial band [25].

5G New Radio (NR) adopts a flexible numerology framework, where different numerologies correspond to different Subcarrier Spacing (SCS) and symbol periods, resulting in variable transmission delay granularity. 3 subcarriers considered in this thesis are presented in table 2 [18].

Numerology (μ)	Subcarrier Spacing (kHz)	Slot Duration (ms)	Cyclic Prefix Type	Symbol Duration (μ s) (Normal CP)	Applicable Scenarios
0	15	1.0	Normal	≈ 66.67	eMBB, mMTC
1	30	0.5	Normal	≈ 33.33	mMTC, moderate URLLC
2	60	0.25	Normal	≈ 16.67	URLLC, low latency

Table 2: Flexible Time—frequency resource structure in 5G NR

See Table 2, these options enable flexible scheduling capabilities from milliseconds to sub-milliseconds intervals, ideal for heterogeneous devices. In industrial deployments, URLLC services typically demand end-to-end latencies below 1 ms. Therefore, configurations like $\mu = 2$ ($\Delta f = 60$) combined with shorter symbol durations are especially suited for mission-critical tasks such as robotic actuation. Conversely, mMTC services may benefit from lower SCS ($\mu=0$), which improves spectral efficiency and supports massive device connectivity with relaxed delay constraints.

To meet the strict end-to-end latency requirements of URLLC devices, this thesis adopts the mini-slot scheduling mechanism [26]. In the case of $\Delta f = 60$ kHz configuration, a two symbol mini-slot with a duration of 16.67 μ s can be used to complete the transmission of downlink packets, which can satisfy the constraints of 1 ms. For some services with slightly higher delay tolerance, ($\Delta f = 15$ kHz, or 30 kHz configurations), are preferable as they can further improve resource utilization efficiency.

To unify the resource allocation model, this thesis defines the basic unit of scheduling as RB, which corresponds to a fixed number of subcarriers in the frequency domain and different mini-slot or slot durations according to the numerology in the time domain. For example, an RB at 15 kHz is about 12 subcarriers $\times$ 0.5 ms, while at 60 kHz it is 12 subcarriers $\times$ 0.25 ms. Let the total available RBs in each 1 ms scheduling be denoted as the set $\mathcal{R}$. The number of RBs in $\mathcal{R}$ depends on the selected numerology configuration and the total bandwidth B_{tot} . These RBs form the total resource pool for downlink scheduling.

3.2 Problem formulation

We tackle the cooperative scheduling challenges of URLLC and mMTC devices with limited wireless resources by creating a multi-objective optimization framework. Extending the approach in [15], this joint optimization not only minimizes total bandwidth usage but also incorporates average communication latency per device, ensuring that each device's QoS requirements are met.

To balance the trade-off between bandwidth efficiency and latency performance, we formulate a weighted multi-objective optimization. Let $\alpha \in [0,1]$ and $\beta = 1 - \alpha$ in which α and β are weighting parameters to achieve a balanced optimization of resource utilization efficiency and latency performance, respectively. Both latency and bandwidth usage are normalized to ensure fairness and prevent one objective from dominating the other. The overall problem is mathematically defined as optimization problem $\mathcal{P}$:

$$\min \quad \alpha \sum_{n=1}^{n_{\max}} z_n f_n + \beta L_{avg} \quad (7)$$

$$\text{subject to} \quad \sum_{n=1}^{n_{\max}} x_{in} = 1, \quad \forall i \in I \quad (7a)$$

$$\sum_{i \in I} x_{in} \leq |I| z_n, \quad \forall n \in [1, n_{\max}] \quad (7b)$$

$$\tau_n + \sum_{l \in H_i} [p_i/p_l] \tau_n \leq L_i, \forall i \in I, \forall n \in [1, n_{\max}] \quad (7c)$$

$$SNR_i \geq SNR_{\text{threshold}}, \forall i \in I_{\text{candidate}} \quad (7d)$$

$$\text{variables} \quad x_{in} \in \{0,1\}, \quad \forall i \in I, \forall n \in [1, n_{\max}] \quad (7e)$$

$$z_n \in \{0,1\}, \quad \forall n \in [1, n_{\max}] \quad (7f)$$

$$f_n \in \{f^1, f^2, \dots, f^{|K|}\} \quad \forall n \in [1, n_{\max}] \quad (7g)$$

The objective function in (7) minimizes total bandwidth usage and average latency across the scheduled device set I . Bandwidth usage is measured by the number of RBs allocated, where each device is assigned one RB from a set of bands $n \in \{1, \dots, n_{\max}\}$ [18], each linked to a numerology type $\mu \in \{0,1,2\}$. A binary variable z_n indicates whether any mMTC device uses band n , and f_n represents bandwidth per resource block for selecting band, calculated by $f_n = 12 \times \Delta f$ [18]. The second component, L_{avg} , captures the average latency across scheduled devices, incorporating both transmission and queuing delays. This term is optimized to meet the stringent QoS requirements of URLLC services, which can be defined as:

$$L_{avg} = \frac{1}{|I|} \sum_{i \in I} \sum_{n=1}^{n_{\max}} x_{in} \cdot \left(\tau_n + \sum_{l \in H_i} \left[\frac{p_i}{p_l} \right] \tau_n \right) \quad (8)$$

where $|I|$ denotes the total number of devices that have successfully received resource allocation in device I . Each device $i_n \in I$ is characterized

by a fixed packet generation period p_i . H_i denotes the set of devices with higher priority than device i and τ_n represents the latency for data transmission. And p_i is the packet generation period of H_i , which is used to calculate the queueing delay.

To ensure practical applicability, the optimization problem is subject to a set of constraints. Constraint (7a) specifies that a device can only be assigned to one band. x_{in} is a binary variable, if $x_{in} = 1$, it means device i has been assigned to band n , so it cannot be assigned again. Constraint (7b) states that if at least one device is assigned to a band, the band must be enabled. Moreover, the total latency of a device L_{total} , comprises queueing delay and transmission delay. Therefore, constraint (7c) limits the total latency of each assigned device i ensuring it does not exceed the maximum latency requirement L_i of its QoS class. Considering the actual quality of wireless channel, constraint (7d) ensures channel condition by requiring the SNR of all scheduled devices to meet a predetermined threshold $SNR_{threshold}$. To ensure the delay sensitivity of critical services during resource allocation, we propose the priority service rule as an implicit constraint. This rule mandates that URLLC devices (Priority=1) are serviced prior to mMTC devices (Priority=0).

This issue falls under the category of mixed-integer nonlinear programming (MINLP) because it involves discrete binary variables z_n and x_{in} , alongside nonlinear latency calculations. Recognized as NP-hard, MINLP problems do not have a polynomial-time optimal solution. Thus, this project demonstrates the effectiveness of the proposed algorithm by verifying that both latency and bandwidth requirements meet QoS requirements.

3.3 Solution to the Multi-Objective Optimization Problem

Given that the original problem is NP-hard due to binary scheduling variables and nonlinear delay constraints, we propose a heuristic algorithm based on service prioritization and channel quality-aware resource allocation. Compared with the existing work, unlike existing work [18], our approach introduces a novel SNR boosting mechanism and multi-numerology mapping strategy for low-latency, high-reliability URLLC services, while maximizing bandwidth efficiency and maintaining fairness for mMTC devices.

The algorithm flowchart is illustrated in Figure2, shows a context-aware scheduling framework designed to manage the coexistence of URLLC and mMTC devices within a unified industrial wireless network. The process begins by inputting the device set, initial transmit power, and an SNR threshold. Devices are first classified based on service type, with URLLC assigned higher priority to reflect their stringent latency and reliability demands. Subsequently, the algorithm determines whether each device operates under LOS or NLOS

conditions, which directly influences channel quality. Initial SNR values are then computed, and for URLLC devices operating in NLOS environments with sub-threshold SNR, targeted SNR boosting is applied to ensure their QoS requirements can be met. Once preprocessing is complete, the algorithm identifies eligible devices for a two-stage multi-numerology resource allocation procedure: high-priority URLLC devices are scheduled first using latency-optimized numerologies, followed by lower-priority mMTC devices scheduled with spectrum-efficient numerologies. The final output includes system-level metrics such as bandwidth utilization and end-to-end latency, enabling scalable and QoS-aware resource scheduling in dynamic industrial environments. We divide the implementation of the algorithm into three phases according to the algorithm flow.

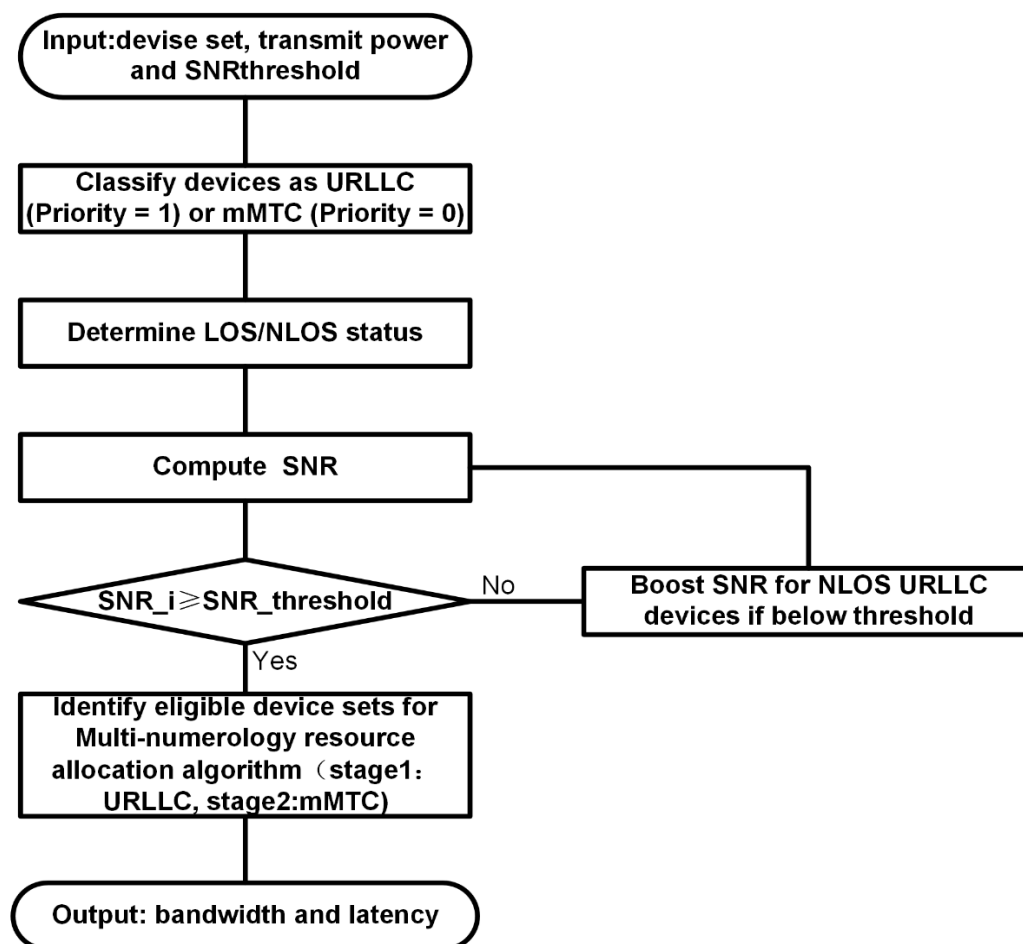


Figure 2: Algorithm Flowchart

3.3.1 Initialization Phase

During the algorithm's initialization phase, we classify devices and initialize wireless resources to prepare for SNR estimation and resource allocation. Initially, the system traverses all available frequency bands (or Numerology configurations), initializing the count of available resource blocks on each band

Total_RB[n] and resetting the counts of used resource blocks RB_Used[n] and corresponding bandwidth usage Used_{BW}[n] to zero (Line 1-4). Subsequently, priority classification is performed: devices identified as URLLC are assigned $Priority = 1$, while others are marked as low priority ($Priority = 0$), directly influencing the resource allocation sequence in the scheduling phase (Line 5-8).

Next, we identify the channel condition for devices in $DeviceList_{URLLC}$ and $DeviceList_{mMTC}$, using the LOS probability $\Pr_{LOS,subscce}$. And save this state as $deviceLOS[i]$ (Line 9). This information is crucial for subsequent SNR boosting and scheduling decisions.

Algorithm1. Initialization Phase - Joint URLLC/mMTC Resource Allocation

Input: Device set I , device types, LoS probability $\Pr_{LOS,subscce}$

Output: $DeviceList_{URLLC}$ 、 $DeviceList_{mMTC}$ 、 $LOS/NLOS\ status$ 、

```

1  for each band n:
2    Initialize total RBs:  $RB_{Available}[n] \leftarrow Total\_RB[n]$ 
3    Set  $RB\_Used[n] \leftarrow 0$ ,  $Used_{BW}[n] \leftarrow 0$ 
4  end for
5  for each device  $i \in I$ :
6    Determine device type and assign priority:
7    if  $deviceType[i] = URLLC$ , then  $Priority[i] \leftarrow 1$ 
8    else  $Priority[i] \leftarrow 0$ 
9    Determine LOS/NLOS status  $deviceLOS[i]$  based on geometry/shadowing
10 end for

```

3.3.2 SNR Estimation and Boosting Phase

Accurate assessment of device wireless channel quality is essential before resource allocation, enabling the system to enhance resource utilization while ensuring reliability. First, based on its initial transmit power P_{tx_global} , calculate the SNR of each device in the set of devices $DeviceList_{URLLC}$ and $DeviceList_{mMTC}$, and filtering the devices with better channel quality by $SNR_{threshold}$ (Line 3). To ensure the reliability of mission-critical communications, we introduce an SNR boosting strategy (Line 4). Specifically, for URLLC devices i experiencing NLOS conditions and having SNR below $SNR_{threshold}$, the base station increases its transmit power for these devices. This boost, denoted as $P_{tx_boosted}$, aims to increase the number of URLLC devices meeting the $SNR_{threshold}$. The SNR for device i is recalculated using this enhanced transmit power (Line 5). Finally, all devices with SNR values meeting or exceeding $SNR_{threshold}$ are

selected to form the candidate set $I_{\text{candidate}}$. This set is then used for subsequent resource scheduling and optimization (Line 6).

Algorithm2. SNR Estimation and Boosting Phase

Initialize device transmit power: $P_{\text{tx,device}}[i] \leftarrow P_{\text{tx,global}}$

Input: $DeviceList_{\text{URLLC}}$, $DeviceList_{\text{mMTC}}$, initial transmit power $P_{\text{tx_global}}$,

$SNR_{\text{threshold}}$

Output: $I_{\text{candidate}}$

for each device $i \in I$:

 Compute initial SNR: $SNR[i] \leftarrow \text{compute_SNR}(P_{\text{tx_device}}[i], i)$

 if deviceLOS[i] = NLOS AND Priority[i] = 1 AND $SNR[i] < SNR_{\text{threshold}}$

 Boost transmit power: $P_{\text{tx_boosted}} \leftarrow P_{\text{tx_global}} + 8 \text{ dB}$

 Recompute SNR: $SNR[i] \leftarrow \text{compute_SNR}(P_{\text{tx_boosted}}, i)$

 Identify eligible device set: $I_{\text{candidate}} \leftarrow \{i \mid SNR[i] \geq SNR_{\text{threshold}}\}$

 end if

end for

3.3.3 Scheduling phase

Following SNR estimation and the construction of the candidate device set $I_{\text{candidate}}$, the resource allocation phase is initiated. Resources are allocated to devices based on their priority and QoS category to ensure that critical services (URLLC) are prioritized to meet low-latency and high-reliability requirements. Resource allocation for URLLC devices is therefore performed first (Stage 1).

Initially, URLLC devices in $DeviceList_{\text{URLLC}}$ that meet the $SNR_{\text{threshold}}$ are sorted by their latency requirements L_i in ascending order to prioritize those with latency sensitivity. Each URLLC device is categorized into one of m QoS classes based on its latency requirement. Each QoS class m corresponds to a preferred numerology μ as defined in a mapping table. During resource allocation, the base station prioritizes these numerologies when allocating resources to devices (Line 2-4).

If available RB remains across all numerologies μ , the Base station allocates one RB to device i and updates the available resources for that numerology. If a device is marked as 'Scheduled', the current scheduling loop for that device is exited (Line 5-10). This phase ensures that URLLC devices receive priority in bandwidth-constrained environments, highlighting the algorithm's delay-sensitive scheduling capability.

Algorithm3. Scheduling phase

Input: $I_{\text{candidate}}$, $\text{DeviceList}_{\text{URLLC}}$, $\text{DeviceList}_{\text{mMTC}}$, QoS classes, initial

transmit power $P_{\text{tx_global}}$, $\text{SNR}_{\text{threshold}}$

Output: Resource allocation results (scheduled devices, RB utilization,)

Stage 1: URLLC Device Scheduling (Priority = 1)

Sort $\text{DeviceList}_{\text{URLLC}}$ by increasing L_i // Delay-sensitive first

for each device $i \in \text{DeviceList}_{\text{URLLC}}$

Got QoS class $m \leftarrow \text{QoS_Class}[i]$

for each numerology $\mu \in \text{MappingTable}[m]$ do

if $\text{RB_Available}[\mu] \geq \text{RB}_i$ then

Allocate RB_i in numerology μ to device i

Update $\text{RB_Available}[\mu] \leftarrow \text{RB_Available}[\mu] - \text{RB}_i$

Update $\text{RB_Used}[\mu] \leftarrow \text{RB_Used}[\mu] + \text{RB}_i$

Mark i as *allocated*

Break

end if

end for

end for

Stage 2: mMTC Device Scheduling (Priority = 0)

Sort $\text{DeviceList}_{\text{mMTC}}$ by increasing L_i

for each device $j \in \text{DeviceList}_{\text{mMTC}}$

Got QoS class $m \leftarrow \text{QoS_Class}[j]$

for each numerology $\mu \in \text{MappingTable}[m]$ do

if $\text{RB_Available}[\mu] \geq \text{RB}_j$ then

Allocate RB_j in numerology μ to device j

Update $\text{RB_Available}[\mu] \leftarrow \text{RB_Available}[\mu] - \text{RB}_j$

Update $\text{RB_Used}[\mu] \leftarrow \text{RB_Used}[\mu] + \text{RB}_j$

Mark j as *allocated*

Break

end if

end for

end for

For mMTC devices, the scheduling process is structurally identical to that of URLLC, and resource allocation also follows its QoS preference numerology mapping table. For simplicity, this part of the scheduling logic is identical to that of the URLLC device, and the detailed repetition of steps is omitted.

4. Results and discussion

To verify whether the algorithm is effective, we need to prove it through simulation. Therefore, in this chapter, we will explain the specific implementation process of the algorithm.

4.1 Simulation Setup

Simulations were conducted using a MATLAB-based system-level model of a 50 m × 50 m industrial factory. A base station was placed 35 m outside the factory center at a height of 5 m, with devices located on the ground. The system operated at a carrier frequency of 3.5 GHz, with a 100 MHz bandwidth and a transmit power of 30 dBm. The user terminal noise figure was set to 9 dB [23]. Antenna gains were configured as −15 dBi for the base station and 3 dBi for the devices.

A total of 100 devices were randomly deployed using Monte Carlo sampling to reflect spatial diversity, with 70% configured as URLLC devices and 30% as mMTC devices. The factory area was divided accordingly to evaluate performance under heterogeneous traffic. To ensure accurate modeling of wireless conditions, LOS and NLOS status were determined for each device, which directly impacted the SNR calculations. In industrial URLLC scenarios, the selection of an appropriate SNR threshold is critical to meet stringent reliability requirements (e.g., $\text{BLER} \leq 10^{-5}$) under varying wireless channel conditions. Simulation-based studies have shown that, in additive white Gaussian noise (AWGN) channels with short-packet transmissions, URLLC links typically require an SNR threshold in the range of 10–16 dB to maintain target reliability for link adaptation [27]. Therefore, the SNR threshold for URLLC reliability assurance was set to 15 dB in this project.

The system supported three flexible numerology configurations: 15 kHz ($\mu = 0$), 30 kHz ($\mu = 1$), and 60 kHz ($\mu = 2$) to reflect the 5G NR and 6G-ready physical layer capabilities. URLLC traffic utilized 2-symbol mini-slots to minimize transmission delay, while mMTC traffic employed 14-symbol normal-slots to maximize spectral efficiency. Slot and symbol durations were derived based on subcarrier spacing and OFDM frame structure, with numerology-dependent bandwidths of 180, 360, and 720 kHz per resource block used for final bandwidth allocation [28].

Furthermore, eight distinct QoS classes were defined to simulate diverse service requirements, each associated with specific latency thresholds and packet arrival periods [10]. An intelligent mapping table was introduced to

associate each QoS class with a preferred numerology, prioritizing higher numerology for latency-sensitive URLLC traffic and lower numerology for bandwidth-efficient mMTC services.

To ensure statistical robustness, 40 Monte Carlo simulation runs were performed per configuration. Metrics such as latency, success rate, and resource utilization were collected for both pre- and post-SNR boosting scenarios. The impact of LOS/NLOS conditions and numerology selection was carefully evaluated for URLLC and mMTC devices across all QoS categories.

4.2 Simulation Results

In this section, we present the simulation results of the algorithm Joint URLLC/mMTC Resource Allocation with the optimization objective described in (7).

4.2.1 Device Deployment and LoS/NLoS Classification

To evaluate the performance of resource allocation strategies in realistic industrial environments, we first simulate the spatial deployment of 100 wireless devices, including both URLLC and mMTC device in a 50 m $\times$ 50 m smart factory. As shown in Figure 3, device locations are generated using Monte Carlo sampling, which enables a statistically robust representation of heterogeneous spatial configurations. This stochastic deployment approach captures the randomness inherent in actual factory floor layouts, reflecting practical variations in user distributions and distances from the base station.

Following the placement, devices are classified as LOS or NLOS based on a distance-dependent probability model as defined by 3GPP standards. Specifically, the LOS probability in 3.1.2 exponentially decays with increasing horizontal distance from the base station, capturing the obstruction effects of the factory environment. The implementation of this model in our simulation (Figure 3) reveals a clear trend: devices located closer to the base station, primarily on the left side of the deployment area, are more likely to maintain LOS conditions (red markers), whereas distant devices tend to experience NLOS links (blue markers).

This classification plays a critical role in subsequent SNR calculations and adaptive resource scheduling. As NLOS links suffer higher PL, their associated devices are more likely to fall below SNR thresholds and therefore require power boosting. The observed LOS/NLOS distribution not only validates the realistic spatial modeling of the environment but also serves as a fundamental input for the proposed SNR-aware dual-stage scheduling mechanism.

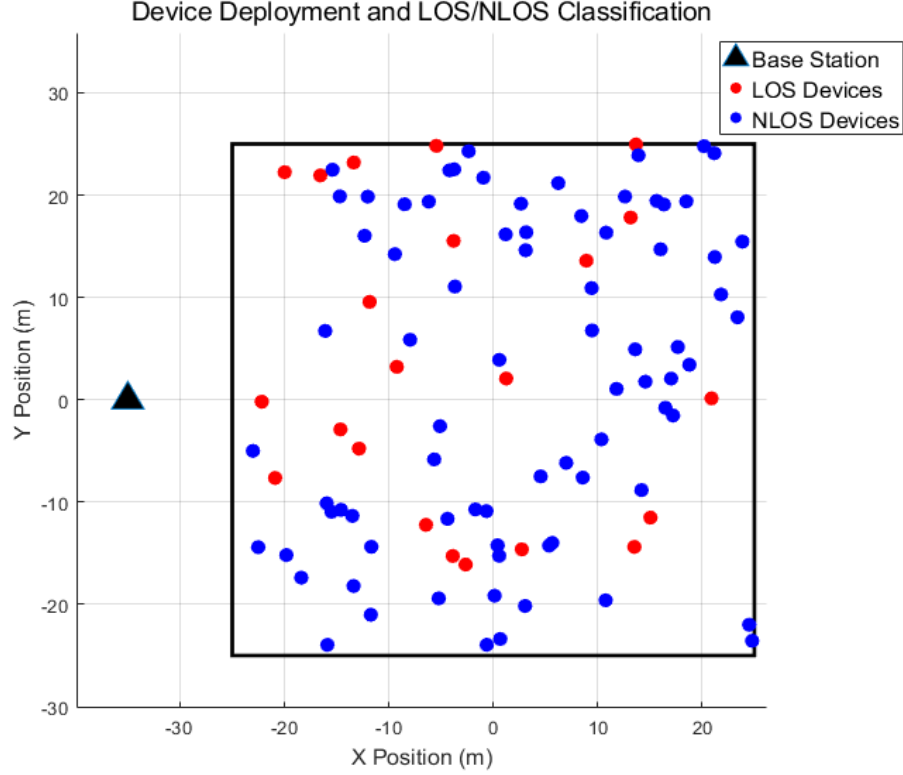


Figure 3: Device Deployment and LOS/NLOS Classification

4.2.2 SNR Distribution before/after Boosting

The SNR of each device was computed using a large-scale fading model, incorporating both PL and shadow fading effects, tailored for industrial propagation environments. For each device, the SNR was calculated based on the equation in 3.1.2. This approach allows accurate modeling of SNR heterogeneity across the factory floor and forms the foundation for analyzing the impact of boosting strategies, as illustrated in the following result figures.

Figure 4 illustrates the SNR distribution (dB) of all devices before and after SNR boosting in the first Monte Carlo simulation, categorized by LOS and NLOS links. In the pre-enhancement scenario (upper subplot), the SNR of some NLOS URLLC devices falls below the 15 dB threshold (indicated by the black dashed line), suggesting that these critical devices struggle to meet communication reliability requirements in the current wireless environment. To enhance the service capability of these devices and ensure their QoS and delay requirements are met, this thesis introduces a priority-based SNR boosting mechanism. The system identifies NLOS URLLC devices (Priority=1) with SNR below the threshold and applies a transmit power boost of 8 dB ($P_{\text{tx_boosted}} = P_{\text{tx_global}} + 8 \text{ dB}$) to these devices. Their enhanced SNR values are then recalculated using the link budget, considering factors such as PL, Shadow Fading, antenna gain, and noise power, in line with 6G channel modeling

requirements.

The post-enhancement SNR distribution (lower subplot) shows that all NLOS URLLC devices (marked by yellow stars) with SNR boosting have successfully exceeded the threshold, qualifying them for subsequent resource allocation. This proactive SNR boosting improves reliability, while the priority elevation ensures critical tasks are always served, enhancing overall system robustness. This SNR boosting strategy is well-suited for factory environments with diverse PL characteristics. It addresses the issue of severe fading in edge NLOS devices due to uneven device distribution and channel conditions. By pre-enhancing the channel quality of key devices, the performance bottleneck can be overcome without significantly increasing system complexity, highlighting the innovation and practical engineering value of this project.

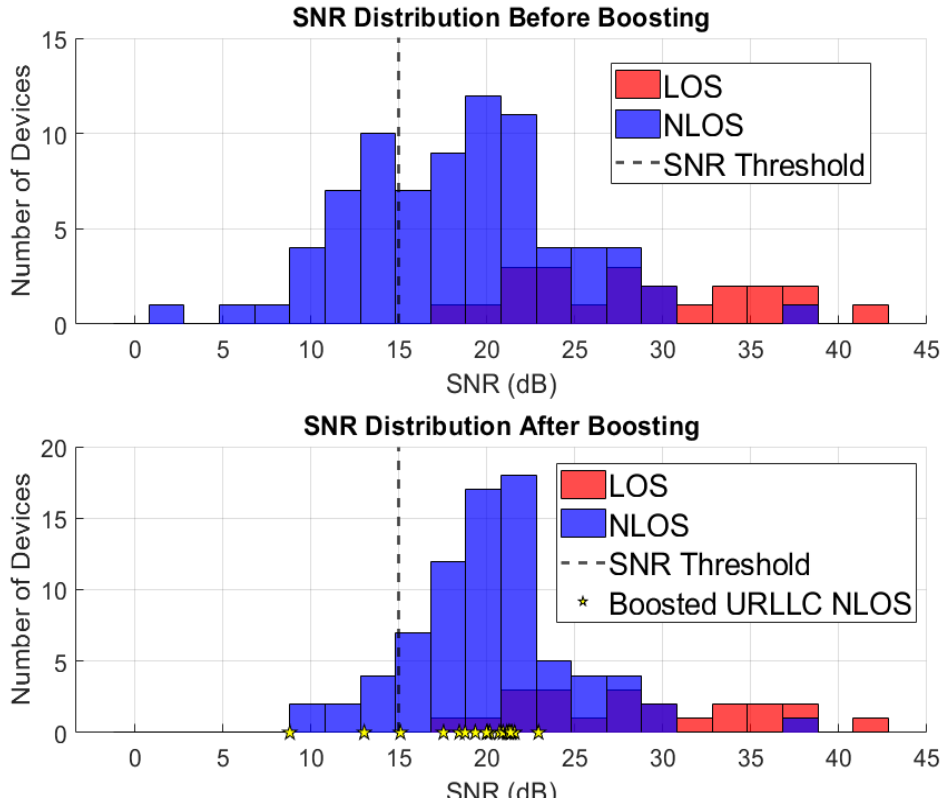


Figure 4: SNR Distribution Before/After Boosting

4.2.3 URLLC Device Latency Distribution

Figure 5 and Figure 6 illustrate the Latency distribution for URLLC devices before and after SNR boosting. Figure 5 represents the end-to-end latency distribution experienced by each successfully allocated URLLC device in these two scenarios, which consists of two main components: queueing delay and transmission delay. The queueing delay accounts for the interference from higher-priority QoS classes that must be served first, considering the packet

generation periods and the number of already allocated devices in the same numerology. In resource allocation for URLLC, we utilize the mini-slot technique and QoS-Numerology mapping, prioritizing delay-sensitive devices to allocate RBs from the 60 kHz numerology to minimize average latency. We use only 2 symbols instead of the standard 14 symbols to achieve ultra-low latency transmission. The average URLLC latency is calculated to quantify the performance improvement achieved by SNR boosting technology, as Figure 5 demonstrates how this technique enhances the low latency of NLOS URLLC devices, thereby increasing the number of devices eligible for resource allocation and ultimately improving the overall latency distribution to meet the stringent 1ms latency requirement of 5G URLLC services. Before SNR boosting, 51 devices met the 1 ms URLLC latency threshold, achieving an average latency of 0.321 ms.

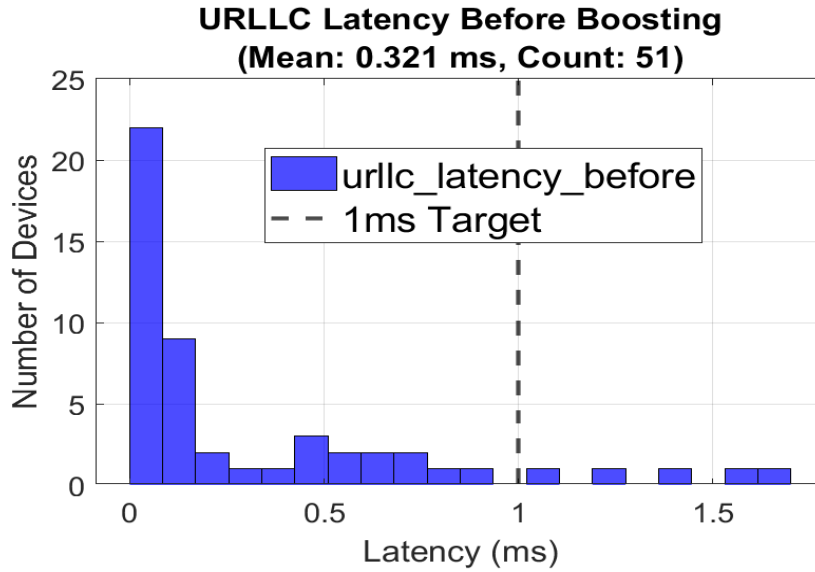


Figure 5: URLLC Latency Before Boosting

Following SNR boosting (Figure 6), the number of compliant devices rose to 68, with the average latency slightly increasing to 0.439 ms. This still satisfies the stringent low-latency requirements of URLLC, which makes our algorithm viable for industrial IoT applications requiring sub-1ms response times. The improvement is mainly due to the SNR boosting process, which enhances the reception quality of devices in weak channels by increasing transmit power. Although the additional devices included in scheduling due to improved SNR threshold compliance have poorer average channel conditions and higher resource scheduling demands, leading to a slight latency increase, this represents a significant performance gain in terms of service reachability and system capacity.

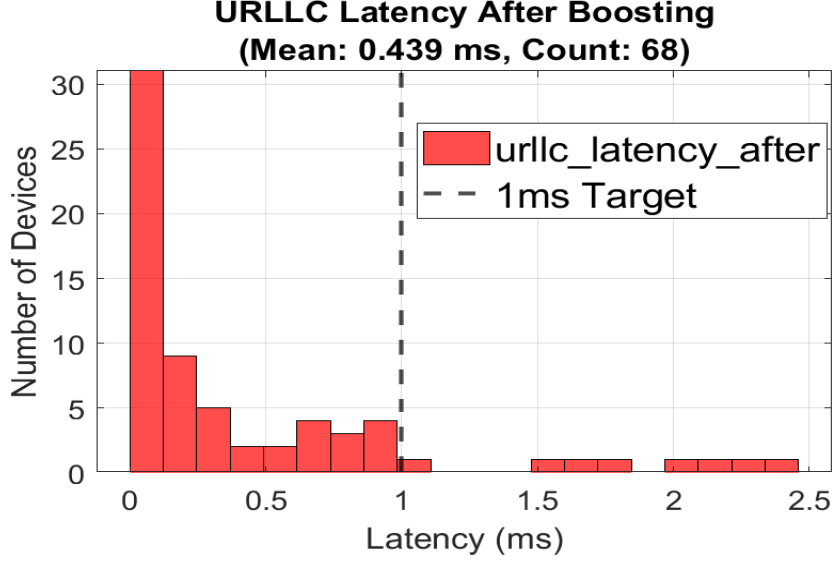


Figure 6: URLLC Latency After Boosting

4.2.4 URLLC Latency CDF Comparison

The URLLC Latency CDF Comparison in Figure 7 serves as a comprehensive statistical visualization tool that demonstrates the cumulative probability distribution of latency performance before and after SNR boosting implementation. Unlike histograms which show frequency distributions, the CDF provides a more informative perspective by revealing the percentage of URLLC devices that achieve latency below any given threshold, making it particularly valuable for assessing compliance with the critical 1ms URLLC requirement. This visualization enables researchers and engineers to quantitatively evaluate the effectiveness of SNR boosting by comparing how the cumulative probability curves shift, showing not only the improvement in mean latency but also enhanced reliability in meeting stringent delay constraints across the entire device population. The CDF format is especially crucial for URLLC applications because it directly answers key performance questions such as "what percentage of devices achieve sub-1ms latency" and allows for precise comparison of tail performance, which is essential for mission-critical applications where even a small percentage of devices exceeding the latency threshold could compromise system reliability.

From Figure 7, it is evident that despite the number of URLLC devices increasing from 51 to 68 post-SNR boosting, and the average delay rising slightly from 0.321 ms to 0.439ms, the system maintains excellent delay control capabilities. The shift in the CDF curve can be attributed to several factors. First, SNR boosting primarily targets URLLC devices in NLOS environments. The high PL and strong shadow fading in NLOS conditions mean some devices require more resources to meet the SNR threshold, potentially increasing their latency. Second, after SNR boosting, the number of schedulable URLLC

devices increases significantly. While this improves coverage, it introduces higher queuing latency due to increased contention for mini-slot resources, leading to a slight increase in overall mean latency. Despite the CDF shifting right, the figure confirms that the proposed scheduling strategy effectively balances SNR and resource optimization. It is well-suited for 6G industrial IoT scenarios demanding high reliability and low latency for URLLC services, demonstrating strong engineering adaptability and potential for practical deployment.

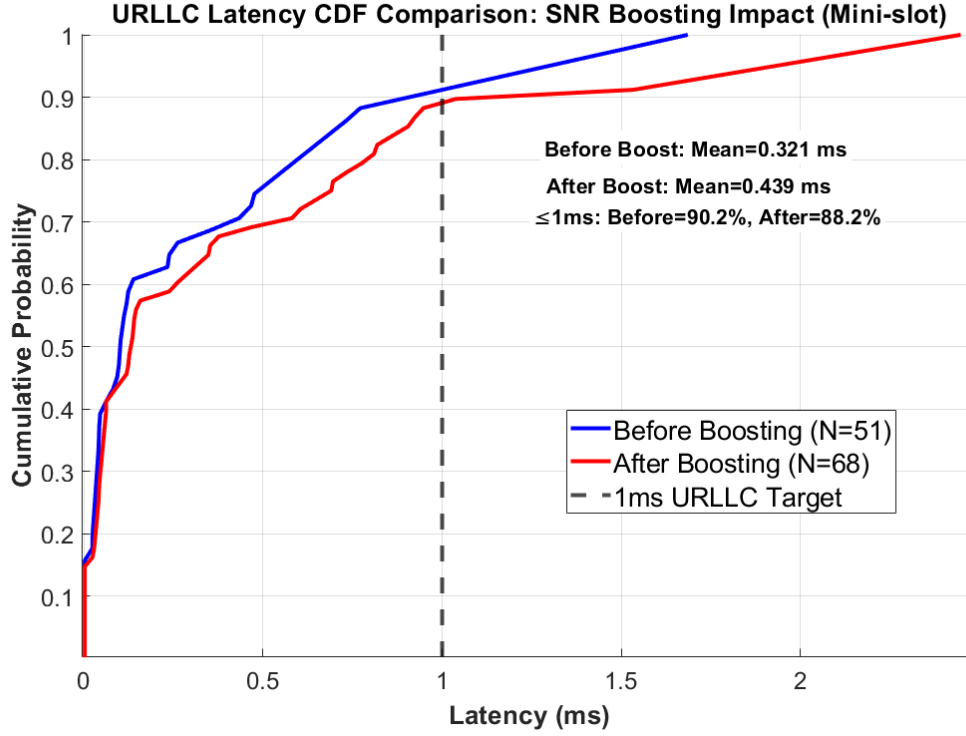


Figure 7: URLLC Latency CDF Comparison

4.2.5 Resource Distribution Results

Figure 8 illustrates the resource distribution results with respect to LOS and NLOS conditions. The left pie chart presents the bandwidth allocation ratio between LOS and NLOS devices, while the right bar chart details the resource allocation pattern, indicating the number of devices successfully allocated or not allocated resources under each channel condition. From the bandwidth allocation chart, it is evident that NLOS devices consume a significantly larger proportion of bandwidth resources compared to LOS devices. This outcome directly reflects the device deployment and LOS/NLOS classification results, where most randomly deployed devices were categorized as NLOS due to distance-dependent propagation characteristics and the exponential LOS probability model. The predominant NLOS allocation pattern stems from the realistic industrial scenario where most IoT devices operate under NLOS conditions due to machinery, structural obstacles, and dense deployment

environments, rather than from any algorithmic bias toward specific channel conditions.

The bar chart on the right further substantiates this observation by showing the number of devices successfully allocated resources. Notably, NLOS devices dominate the allocated group, with over 70 devices receiving resources, while only about 20 LOS devices are scheduled. This distribution is consistent with the underlying device deployment pattern rather than preferential treatment, highlighting the algorithm's fairness in maintaining proportional resource allocation that accurately reflects the actual network topology and propagation environment. This resource allocation pattern arises from the QoS-aware and SNR-driven hierarchical scheduling algorithm employed in this study. The algorithm prioritizes devices based on service type (e.g., URLLC vs. mMTC), channel quality (LOS vs. NLOS), and numerology adaptability. The system achieves equitable resource distribution by ensuring that allocation decisions correspond to the realistic device distribution and channel conditions, thereby ensuring that critical communication needs are met across all propagation environments without artificial bias.

While the preference toward NLOS devices may seem resource-intensive, it reflects the algorithm's design goal of maintaining fairness and realistic resource provisioning in imbalanced channel conditions. The allocation pattern demonstrates the system's ability to serve most devices operating under challenging NLOS conditions while efficiently accommodating LOS devices, ensuring comprehensive coverage and operational reliability. Importantly, the results demonstrate the system's capacity to balance QoS provisioning and spectrum efficiency, accommodating a diverse range of devices without overwhelming the system.

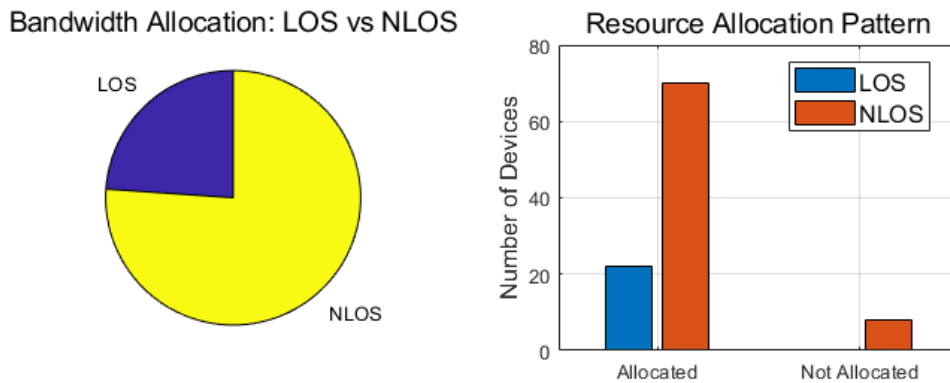


Figure 8: URLLC Latency CDF Comparison

4.2.6 Resource Utilization of URLLC and mMTC Device

Figure 9 compares bandwidth utilization for different service types (URLLC and mMTC) and propagation conditions (LOS and NLOS), highlighting the influence of slot structure and environmental variability on system efficiency. The system adopts a differentiated slot allocation strategy. URLLC device is scheduled using 2-symbol mini-slots to meet strict latency and reliability requirements, whereas mMTC devices utilize 14-symbol normal slots to enhance spectral efficiency for delay-tolerant communications.

As illustrated in the figure, URLLC devices under NLOS conditions exhibit the highest bandwidth utilization, achieving approximately 5%. This outperforms both their LOS counterparts ($\approx 2\%$) and all mMTC configurations, which achieve only $\sim 2.1\%$ (NLOS) and $\sim 0.5\%$ (LOS), respectively. This result demonstrates the system's effective prioritization of URLLC traffic, particularly in challenging NLOS scenarios where reliable connectivity is harder to maintain. The SNR boosting mechanism (as discussed earlier) ensures that URLLC devices with severe attenuation are successfully elevated above the SNR threshold, making them eligible for resource scheduling. Consequently, a greater proportion of resources is actively utilized by these high-priority devices, contributing to the elevated bandwidth utilization observed. The noticeably lower utilization by mMTC devices stems from their lower scheduling priority and the use of larger slot durations, which can lead to underutilization when traffic demand is sparse or when channel conditions hinder reliable access. Additionally, since mMTC devices are more tolerant of delay and have less stringent QoS requirements, the algorithm naturally allocates fewer time-frequency resources to them in favor of latency-sensitive URLLC traffic.

These findings affirm the design efficacy of the proposed multi-numerology resource allocation scheme. By adaptively allocating mini- or normal-slots based on both device priority and link quality (LOS/NLOS classification), the algorithm dynamically balances system efficiency with QoS guarantees. This strategy is particularly beneficial in industrial 6G deployments, where heterogeneous service demands coexist in highly variable radio environments.

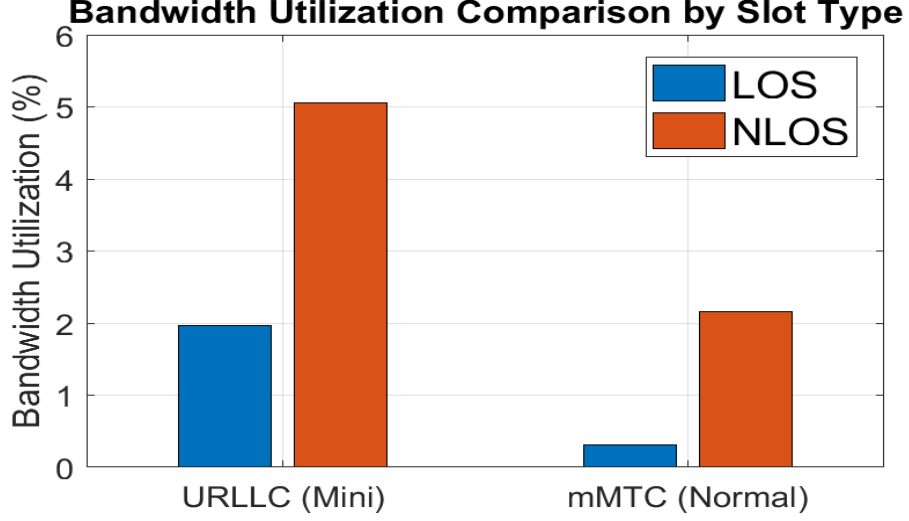


Figure 9: Bandwidth Utilization Comparison by Slot Type

Figure 10 evaluates the overall resource allocation efficiency across different slot configurations and propagation conditions by comparing URLLC (mini-slot) and mMTC (normal-slot) devices under LOS and NLOS scenarios. This metric is computed by classifying all devices into four categories and calculating the ratio of devices in each category that successfully received resource allocation to the total number of devices within that category. This classification and evaluation process enables a fine-grained analysis of the system's behavior under heterogeneous service demands and wireless propagation environments.

The results reveal that URLLC devices consistently achieve near-perfect allocation efficiency (100%) in both LOS and NLOS conditions, reflecting the effectiveness of the proposed priority-based scheduling mechanism in guaranteeing service for ultra-reliable, low-latency traffic. This is largely attributable to two key mechanisms: first, the hierarchical resource allocation algorithm prioritizes URLLC devices during the first scheduling stage; and second, the integrated SNR boosting procedure ensures that even NLOS URLLC devices, which typically suffer from poor channel quality, are elevated above the SNR threshold and become eligible for scheduling.

In contrast, mMTC devices exhibit lower resource allocation efficiency, particularly in NLOS environments, where their efficiency drops to approximately 74%. While mMTC-LOS devices still benefit from stable link quality and achieve relatively high success rates, their overall performance is bounded by their secondary scheduling priority and the lack of SNR boosting. This outcome is expected, as mMTC traffic is designed for massive connectivity with relaxed latency constraints, allowing the algorithm to conserve system resources by limiting their allocation under adverse conditions.

The observed efficiency trends validate the adaptability and fairness of the proposed scheduling framework. By leveraging slot type (mini vs. normal) and

channel classification (LOS vs. NLOS), the algorithm dynamically allocates resources to meet diverse QoS requirements. It ensures critical low-latency services are reliably supported without excessively sacrificing the performance of delay-tolerant services. Moreover, this result underscores the practicality of the solution in 6G industrial IoT scenarios, where reliable real-time control and scalable sensor connectivity must coexist within the same wireless infrastructure.

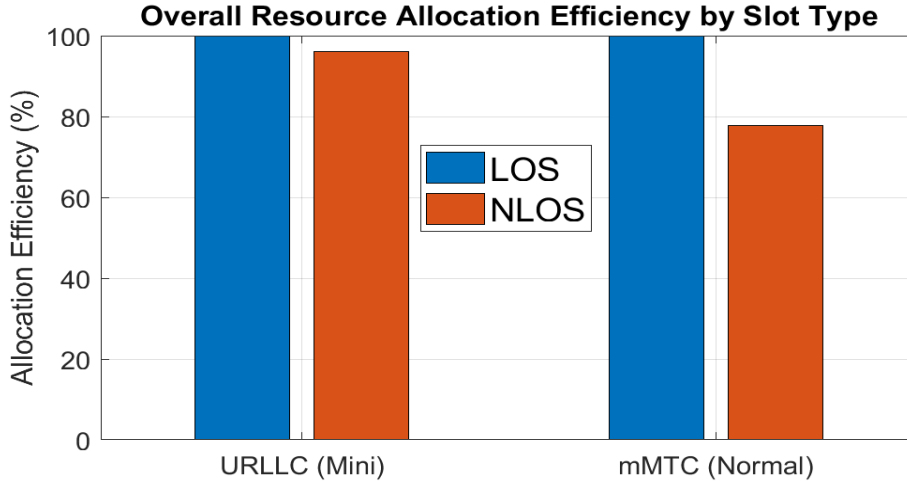


Figure 10: Overall Resource Allocation Efficiency by Slot Type

4.2.7 SNR Boosting Improvement per MC Run

In wireless communication systems, stochastic factors such as device location, PL, and shadow fading significantly impact system performance, making it challenging for a single simulation to comprehensively reflect overall performance. To assess the robustness of the proposed scheduling and SNR boosting algorithm across different stochastic scenarios, we conducted 40 independent Monte Carlo simulations, varying conditions such as device distribution and PL in each run, to statistically evaluate the algorithm's stability.

Figure 11 presents the statistical distribution of SNR boosting gains across 40 independent Monte Carlo simulation runs, focusing on the number of additional URLLC devices that successfully meet the SNR threshold after the boosting mechanism is applied. Each Monte Carlo run involves a random deployment of devices within the factory environment, reflecting realistic spatial variability and heterogeneous wireless conditions. This stochastic setup is essential for evaluating the robustness and generalizability of the SNR boosting strategy under practical industrial scenarios.

In each run, the system identifies NLOS URLLC devices whose initial SNR falls below the critical threshold of 15 dB, rendering them ineligible for reliable resource allocation, and then selectively applies an 8 dB transmit power boost.

This process effectively raises the signal strength of edge-case devices, enabling them to overcome high PL and shadow fading effects common in industrial NLOS conditions. The number of successfully recovered devices is then recorded per run to assess the consistency and impact of the boosting mechanism.

The results show an average improvement of 14.4 devices per run, with a standard deviation of ± 2.6 devices, corresponding to a mean improvement rate of 20.6% over the baseline scenario without boosting. The relatively low variance across the 40 runs confirms that the proposed enhancement strategy performs consistently, despite the randomness in device placement and channel conditions. These findings suggest that the SNR boosting mechanism is not only effective but also statistically reliable across a wide range of spatial and propagation conditions. From a system design perspective, this level of improvement is significant. By enabling more URLLC devices to qualify for resource allocation, the boosting strategy directly enhances the network's capability to support latency-critical industrial applications, such as real-time control and safety monitoring. Importantly, the power increase is selectively applied only to critical NLOS URLLC devices, thereby minimizing unnecessary interference and preserving energy efficiency is an essential consideration in dense industrial deployments.

Overall, Figure 11 highlights the algorithm's capacity to proactively adapt to channel imperfections and recover critical connectivity without requiring extensive changes to scheduling complexity or infrastructure. Statistical evidence reinforces the universal applicability of the SNR boosting strategy and underscores its value in supporting robust and reliable URLLC services in 6G industrial IoT networks.

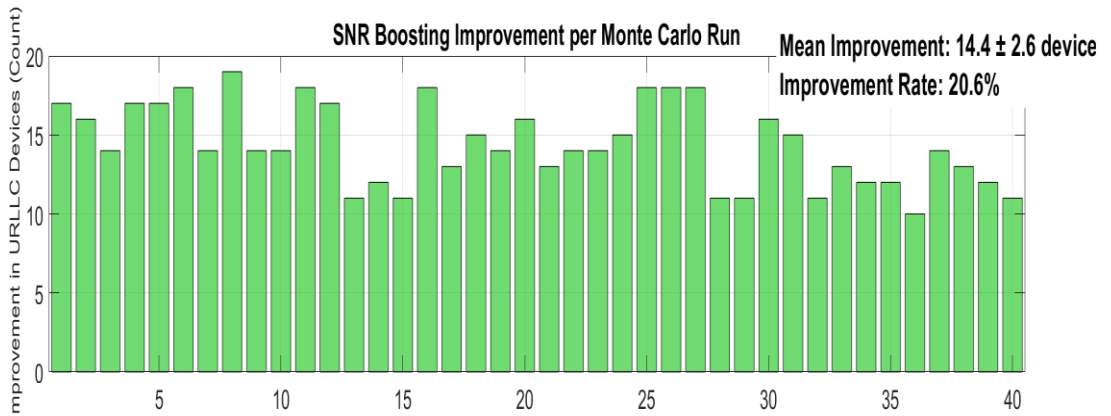


Figure 11: SNR Boosting Improvement per Monte Carlo Run

4.2.8 URLLC Latency Evolution Across 40 Monte Carlo Runs

Figure 12 illustrates the evolution of average URLLC latency across 40 Monte Carlo simulation runs, comparing system performance before and after SNR boosting under the mini-slot configuration. This analysis provides critical insights into how the proposed SNR boosting strategy affects latency-sensitive URLLC services under realistic and stochastic deployment conditions. Each Monte Carlo run generates a unique spatial distribution of devices and channel characteristics, ensuring statistical robustness of the results. The average latency is computed by aggregating delay values for all successfully scheduled URLLC devices within each run. The use of 2-symbol mini-slots, as mandated for URLLC traffic, enables low-latency transmissions, thereby serving as an ideal testbed for evaluating the proposed strategy's impact on timing performance.

Before the application of SNR boosting, the average latency across all runs was 0.426 ± 0.121 ms, well within the sub-millisecond URLLC requirement. Following SNR boosting, the latency increased to 0.545 ± 0.138 ms, as more NLOS URLLC devices with previously insufficient SNR were brought into the scheduling pool. This marginal increase is attributed to a higher number of devices competing for mini-slot resources after being rescued by the boosting mechanism, leading to slightly longer queuing or scheduling delays.

Despite this increase, the average latency consistently remains well below the 1 ms threshold, thus fully satisfying the strict delay constraints of industrial URLLC use cases. Importantly, the post-boosting latency variation remains controlled across all 40 runs, indicating that the system maintains performance stability even under varying channel conditions and deployment topologies. These findings validate the effectiveness of the SNR boosting scheme not only in improving coverage and reliability (as shown in previous figures), but also in preserving latency performance under increased network load. By intelligently incorporating previously unqualified URLLC devices into the scheduling process without significantly sacrificing delay performance, the algorithm enhances the overall system utility while meeting critical QoS guarantees. This result demonstrates that the proposed approach achieves a favorable trade-off between reliability improvement and latency preservation, making it well-suited for URLLC applications in future 6G industrial networks.

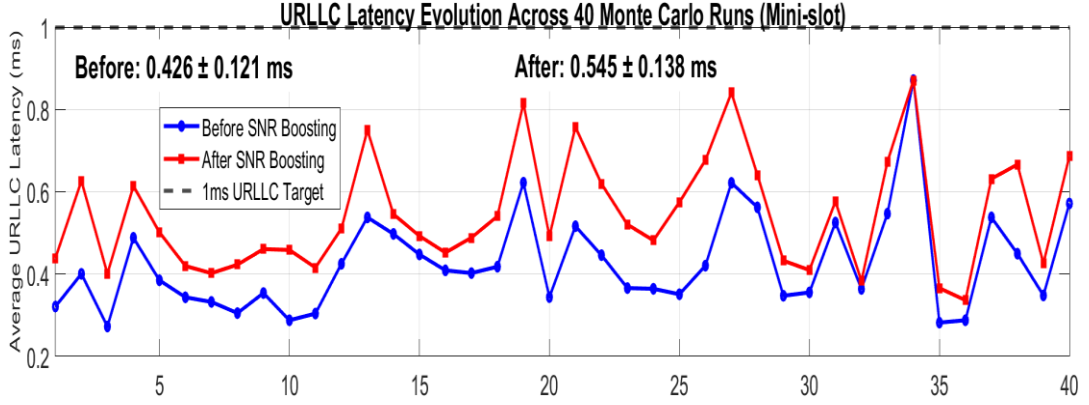


Figure 12: URLLC Latency Evolution Across 40 Monte Carlo Runs

4.2.9 Service Type Resource Utilization Comparison with Trends

Figure 13 presents a comparative analysis of resource utilization between URLLC and mMTC services across 40 Monte Carlo simulation runs, offering insights into the performance stability and prioritization efficacy of the proposed scheduling framework. The simulation considers dynamic variations in device locations, propagation conditions, and random network configurations, ensuring comprehensive statistical validity.

As shown in Figure 13, URLLC devices consistently utilized $74.8\% \pm 2.3\%$ of the available resources, while mMTC devices accounted for $25.2\% \pm 2.3\%$, yielding an average URLLC to mMTC utilization ratio of approximately 2.96. This utilization pattern is closely aligned with the system's device composition, which reflects realistic industrial traffic demands. The observed ratio validates that the proposed algorithm honors service-level proportionality while dynamically adapting to link quality variations and QoS demands. The trend lines superimposed on the plot further emphasize the system's low variance and consistent behavior over multiple random trials. URLLC resource consumption remains slightly above its proportionate share, which can be attributed to the algorithm's prioritization of ultra-reliable, low-latency traffic. This prioritization ensures that mission-critical applications receive preferential scheduling, particularly in adverse channel environments or contention scenarios.

The algorithm achieves this prioritization using a QoS-aware heuristic that integrates service type, SNR, and numerology selection into a hierarchical allocation process. This design allows the system to dynamically respond to mixed service demands without sacrificing fairness or stability. Importantly, despite URLLC's higher allocation, mMTC devices consistently receive a stable and sufficient share of resources, ensuring their massive connectivity needs are not compromised. These results in Figure 13 demonstrate that the proposed

scheduling algorithm achieves a balanced trade-off between service differentiation and fairness, ensuring reliable URLLC performance while sustaining mMTC accessibility. The system's robustness across diverse Monte Carlo realizations confirms its applicability in real-world 6G industrial IoT environments, where predictable resource provisioning under dynamic traffic conditions is paramount.

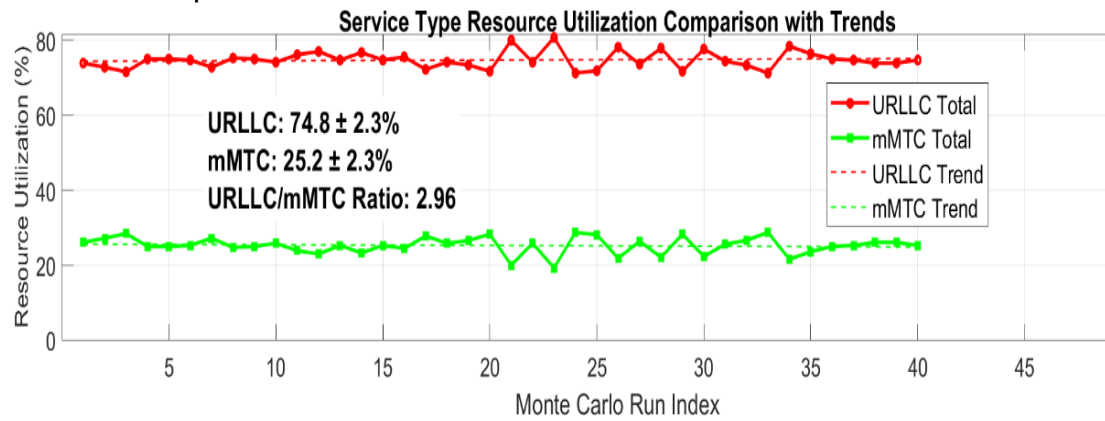


Figure 13: Service Type Resource Utilization Comparison with Trends

5. Conclusion

This project presents a novel and practical resource allocation framework tailored for the coexistence of URLLC and mMTC services in 6G-enabled smart factory environments, providing a viable and scalable solution for heterogeneous QoS support in future 6G industrial IoT deployments. By integrating flexible numerology selection, slot type adaptation, and QoS-aware priority control at the physical layer, the proposed approach addresses key challenges arising from heterogeneous device requirements, delay constraints, and varying channel conditions. A low-complexity heuristic scheduling algorithm is employed to efficiently allocate radio resources while minimizing computational overhead. To further enhance reliability for mission-critical communications, especially in adverse NLOS conditions, a priority-based SNR boosting mechanism is introduced. This mechanism proactively identifies URLLC devices suffering from low SNR and selectively boosts their transmission power, ensuring their inclusion in resource allocation despite unfavorable propagation conditions.

Comprehensive simulation results validate the effectiveness of the proposed framework. Simulation results show a 20.6% increase in the number of URLLC devices meeting scheduling requirements, with average latency maintained at 0.439ms within industrial real-time limits. Moreover, the resource allocation efficiency reached nearly 100% for URLLC and 75% for mMTC devices, indicating optimal utilization of available bandwidth. Bandwidth and resource utilization trends further confirm that the algorithm maintains a balanced prioritization strategy, favoring high-QoS services without sacrificing fairness or scalability. This resource allocation efficiency metric offers a comprehensive lens to assess scheduling performance under compound QoS and channel constraints. It demonstrates that the proposed algorithm can not only maximize utilization but also maintain equitable treatment across service types, affirming its robustness and practical relevance for future wireless industrial networks.

The overall results underscore the algorithm's ability to operate stably and adaptively across diverse Monte Carlo simulations, offering consistent performance under stochastic conditions. The trend-based analysis confirms the algorithm's resilience to dynamic channel fluctuations, while the mini-slot-based scheduling for URLLC services enables low-latency communication without compromising spectrum efficiency.

Future work could explore incorporating reinforcement learning for optimization and extending the framework to multi-node cooperative scheduling architectures, thereby enhancing system intelligence and scalability.

6. Reference

- [1] A. Hazra, A. Munusamy, M. Adhikari, and L. K. Awasthi, "6G-Enabled Ultra-Reliable Low Latency Communication for Industry 5.0: Challenges and Future Directions," *IEEE Communications Standards Magazine*, vol. 8, no. 2, pp. 36-42, 2024.
- [2] M. Soori, B. Arezoo, and R. Dastres, "Internet of Things for smart factories in industry 4.0, a review," *Internet of Things and Cyber-Physical Systems*, vol. 3, pp. 192-204, 2023.
- [3] W. Xia, J. Goh, C. A. Cortes, Y. Lu, and X. Xu, "Decentralized coordination of autonomous AGVs for flexible factory automation in the context of Industry 4.0," in *2020 IEEE 16th International Conference on Automation Science and Engineering (CASE)*, 2020: IEEE, pp. 488-493.
- [4] S. R. Pokhrel, J. Ding, J. Park, O.-S. Park, and J. Choi, "Towards enabling critical mMTC: A review of URLLC within mMTC," *IEEE access*, vol. 8, pp. 131796-131813, 2020.
- [5] I. WP5D, "Future technology trends of terrestrial international mobile telecommunications systems towards 2030 and beyond," *International Telecommunication Union, Report M*, pp. 2516-0, 2022.
- [6] P. K. Verma et al., "Machine-to-Machine (M2M) communications: A survey," *Journal of Network and Computer Applications*, vol. 66, pp. 83-105, 2016.
- [7] J. Bilbao, C. Cruces, and I. Armendariz, "Methodology for the QoS Characterization in High Constraints Industrial Networks," *Open Journal of Communications and Software*, vol. 1, no. 1, pp. 30-41, 2014.
- [8] R. N. Gore, E. Lisova, J. Åkerberg, and M. Björkman, "Delay and Jitter Analysis in Industrial Control Systems: A Paper Mill Case Study," in *2021 17th IEEE International Conference on Factory Communication Systems (WFCS)*, 2021: IEEE, pp. 99-106.
- [9] D. Xia, J. Wan, P. Xu, and J. Tan, "Deep reinforcement learning-based QoS optimization for software-defined factory heterogeneous networks," *IEEE*

Transactions on Network and Service Management, vol. 19, no. 4, pp. 4058-4068, 2022.

[10] A. Mahmood, M. I. Ashraf, M. Gidlund, and J. Torsner, "Over-the-air time synchronization for URLLC: Requirements, challenges and possible enablers," in *2018 15th International Symposium on Wireless Communication Systems (ISWCS)*, 2018: IEEE, pp. 1-6.

[11] B. Adhikari, M. Jaseemuddin, and A. Anpalagan, "Resource allocation for co-existence of embb and urllc services in 6g wireless networks: A survey," *IEEE Access*, vol. 12, pp. 552-581, 2023.

[12] A. Yazar and H. Arslan, "A flexibility metric and optimization methods for mixed numerologies in 5G and beyond," *IEEE Access*, vol. 6, pp. 3755-3764, 2018.

[13] A. Mirri, D. Forlivesi, R. Schiavone, R. Garelo, M. Chiani, and E. Paolini, "A Flexible Scheme for Critical mMTC," in *2024 IEEE 8th Forum on Research and Technologies for Society and Industry Innovation (RTSI)*, 2024: IEEE, pp. 426-431.

[14] A. E. Kalor, G. Durisi, S. Coleri, S. Parkvall, W. Yu, A. Mueller, and P. Popovski, "Wireless 6g connectivity for massive number of devices and critical services," *Proceedings of the IEEE*, 2024.

[15] H. L. dos Santos, J. C. Marinello Filho, C. M. Panazio, and T. Abrão, "Random Access Based on LSTM for Mixed Traffic IoT Networks," *IEEE Transactions on Network Science and Engineering*, vol. 11, no. 3, pp. 2561-2573, 2023.

[16] J. Li, H. Wu, Y. Niu, B. Ai, N. Wang, and T. Q. Quek, "Resource allocation for downlink URLLC in a smart factory," in *ICC 2024-IEEE International Conference on Communications*, 2024: IEEE, pp. 605-610.

[17] M. Sharara, A. M. Masucci, Y. Carlinet, and N. Perrot, "Optimizing URLLC Resource Allocation in Open-RAN: A Transformer-based Approach to Flexible Numerology," in *IEEE INFOCOM 2025-IEEE Conference on Computer*

Communications Workshops (INFOCOM WKSHPS), 2025.

[18] Y. Sadi, S. Erkucuk, and E. Panayirci, "Flexible physical layer based resource allocation for machine type communications towards 6G," in *2020 2nd 6G Wireless Summit (6G SUMMIT)*, 2020: IEEE, pp. 1-5.

[19] A. Yazar and H. Arslan, "Flexible multi-numerology systems for 5G new radio," *Journal of Mobile Multimedia*, vol. 14, no. 4, pp. 367-394, 2018.

[20] L. Marijanovic, S. Schwarz, and M. Rupp, "Multi-user resource allocation for low latency communications based on mixed numerology," in *2019 IEEE 90th Vehicular Technology Conference (VTC2019-Fall)*, 2019: IEEE, pp. 1-7.

[21] G. S. Kesava and N. B. Mehta, "Multi-connectivity for URLLC and coexistence with eMBB in time-varying and frequency-selective fading channels," *IEEE Transactions on Wireless Communications*, vol. 22, no. 6, pp. 3599-3611, 2022.

[22] "3rd Generation Partnership Project; Technical Specification Group Radio Access Network; NR; Base Station (BS) radio transmission and reception (Release 17)," 3rd Generation Partnership Project (3GPP), 3GPP Technical Specification TS 38.104, 2024, 3GPP TS 38.104 v17.18.0 Release 17. [Online]. Available: https://www.3gpp.org/ftp/Specs/archive/38_series/38.104/38104-hv0.zip

[23] 3GPP, "Study on channel model for frequencies from 0.5 to 100 ghz," 3rd Generation Partnership Project (3GPP), Technical Report TR 38.901, Mar 2022, release 17. [Online]. Available: https://www.3gpp.org/ftp/Specs/archive/38_series/38.901/

[24] F. Guidi, A. Guerra, and D. Dardari, "Personal mobile radars with millimeter-wave massive arrays for indoor mapping," *IEEE Transactions on Mobile Computing*, vol. 15, no. 6, pp. 1471-1484, 2015.

[25] T. Jiang *et al.*, "3GPP standardized 5G channel model for IIoT scenarios: A survey," *IEEE Internet of Things Journal*, vol. 8, no. 11, pp. 8799-8815, 2021.

[26] A. Yazar and H. Arslan, "Reliability enhancement in multi-numerology-

based 5G new radio using INI-aware scheduling," *EURASIP Journal on Wireless Communications and Networking*, vol. 2019, no. 1, p. 110, 2019.

[27] J. Khan and L. Jacob, "Investigation of 5G NR physical layer enablers for URLLC: a simulation study," *Sāadhanā*, vol. 48, no. 2, p. 77, 2023.

[28] M. Alsenwi, K. Kim, and C. S. Hong, "Radio resource allocation in 5g new radio: A neural networks based approach," *arXiv preprint arXiv:1911.05294*, 2019.